RENJITH.R.L

B.E. Electronics and Communication Engineering Student

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Website: https://rlrenjith71-star.github.io/portfolio/ **|Location: Nagercoil,TamilNadu.**

PROFESSIONAL SUMMARY

Motivated third-year Electronics and Communication Engineering undergraduate specializing in Embedded systems, IoT, and VLSI design. Possesses strong technical skills in hardware development, microcontroller programming, digital electronics, and semiconductor fundamentals. Adept at problem-solving, circuit prototyping, and leading cross-functional project teams. Actively seeking Embedded Systems Engineer, Hardware Engineer, or VLSI Design Engineer internship opportunities to contribute hands-on technical skills and apply industry-relevant technologies.

EDUCATION

- **B.E ELECTRONICS AND COMMUNICATION ENGINEERING** **2024-Present**
  Rohini College of Engineering and Technology.Palkulam.
  CGPA:9.35
- **Minor Degree: VLSI DESIGN** **2026-Present**
  Rohini College of Engineering and Technology, Palkulam.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Embedded C, JavaScript
- **Embedded Systems & IoT:** Microcontrollers(Arduino,ESP32,STM32), Sensors, Actuators
- **VLSI & Electronics:** Digital Electronics, RTL Design, Verilog HDL
- **Core Electronics:** Circuit Design and Debugging
- **Web Development:** HTML, CSS, JavaScript, React, GitHub, Vercel, Netlify
- **Tools:** MATLAB, Proteus, ArduinoIDE, Microsoft Excel, Microsoft Word, PowerPoint, VS Code
- **Soft Skills:** Team Work, Leadership, Communication, Time Management

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INTERNSHIP

MERN Stack Development,PABOK Technologies, Nagercoil **April 2026- June 2026**

- Implemented responsive frontend UI components and structured web interfaces using HTML,CSS, JavaScript, and React.
- Contributed to backend application logic and basic database integration following MERN stack development practices.
- Collaborated with team members on code reviews, debugging, and software feature integration.

PROJECTS

1. **MOSA – Mobile Organic Smart Agriculture Assistant | Team Leader**

   - Developed an IoT-based smart agriculture robot using ESP32, Arduino IDE, C++, sensors, and AI-based vision for automated irrigation, pest control, weed removal, and fruit harvesting.
   - Shortlisted for the final round of Nimirindhu Nil Hackathon '26 (State-level competition).

- Runner-Up & State Finalist: Technova '26 (Madurai Zonal Level).

2. **Guardian SOS App – Secure Application for Safety of People| Team Leader**

    - Directed development of a mobile emergency safety application incorporating live location sharing, route deviation alerts, safety check-in timers, and emergency contacts.
    - Configured automated SMS and Firebase Cloud Messaging alert notifications to optimize response times during critical situations.

3. **Smart Helmet Using ESP32 | Team Leader**

    - Prototyped a safety monitoring helmet integrating ESP32, gas/alcohol sensors, an analog accelerometer, ultrasonic distance sensors, GPS, and a buzzer.
    - Built alert systems to monitor environmental hazards and vehicle motion anomalies in real time.

4. **Smart Pill Dispenser with Voice Reminder System| Team Leader**

    - Built an automated medication reminder prototype leveraging ESP32, a DS3231 RTC module, servo motors, and a DFPlayer Mini.
    - Programmed scheduled compartment unlocking and synchronized audio reminders to improve medication adherence.

5. **Portfolio Website**

CERTIFICATIONS

- MongoDB Basics for Students – MongoDB(July 2025)
- MERN Stack Development – PABOK Technologies (June 2026)

LANGUAGES KNOWN

- Tamil
- English
- Malayalam
